Dr. Vilas K. Chitrakaran, Ph.D., C.Eng. (MIET)

London, UK; web: https://cvilas.github.io/; e-mail: vilas@ieee.org

About

Engineering leader with 20+ years building low-latency, high-performance software for robots and autonomous systems across industrial and consumer domains. Close to a decade of experience building and leading high-performance teams at the intersection of ambiguity, possibility and real-world deployment at scale.

- Pedigree: Clemson (Ph.D. nonlinear systems), Imperial College (Executive Education), GE Aerospace, Saab, Dyson, Arrival
- Leadership philosophy: https://github.com/cvilas/guidance/
- Software craftsmanship: https://github.com/cvilas/grape

Areas of Expertise

- **Product development**: Full lifecycle from concept (TRL1) to product (TRL8).
- **Leadership**: Technology and innovation strategy, people operations, external partnerships
- **Engineering expertise**: Distributed real-time software architectures for industrial robotic machinery; kinematic and dynamic modeling of arms, legs and wheels; algorithms for state estimation and signal processing; design of command and control systems; human-machine interfaces; immersive 3D visualization and simulation systems; machine vision.
- **Technology stack**: Multi-paradigm software design - C++23, Python, (Rust); distributed heterogenous communication, computation and action - Zenoh, DDS, CANOpen, GPGPU; graphical system design - OpenGL, Qt, (Vulkan); OS/RTOS - Linux, QNX; standards - MISRA, JSF AV Rules, ISO26262

Professional Experience

- **Head of Onboard Software, Mobile Robotics** *Sep. 2023 - Jul. 2026*
  All3, London, United Kingdom

  - Established the software infrastructure, built the 'monorepo', wrote many high performance core libraries including logging, scheduling, messaging and signal processing (C++23)
  - Grew a team of about ten direct reports, and established standards and norms covering technical excellence and team culture

- **Software Team Lead/Principal Engineer, Mobile Robotics** *Aug. 2018 - Sep. 2023*
  Arrival, London, United Kingdom

  - Conceived the software stack for Arrival's industrial mobile robotic fleet, a distributed system of hundreds of robotic agents working as one to deliver the vision of flexible (software-defined) manufacturing. (https://youtu.be/G4UGFzaSHw8)
  - Nurtured a team of about ten software engineers that built it.
  - Implemented core components of the stack: device drivers, messaging middleware, teleoperation, telemetry, robot control, and tools for field operations.
  - Established processes to deliver software fortnightly into the production environment.
  - Provided consultancy services to transfer above technology to one of the largest automotive companies on Earth.

- **Senior Systems Engineer** *Apr. 2017 - Aug. 2018*
  Intelligent Robots, London, United Kingdom

  - Developed product concept for an autonomous robotic system for warehouse intra-logistics.
  - Won external funding to develop a prototype for commercial validation.
  - Implemented core software components for the prototype.

- **Senior Robotic Algorithms Engineer** *Sep. 2013 - Mar. 2017*
  Dyson, Malmesbury, United Kingdom

  - Member of the team that delivered the *Dyson 360 Eye* robot, their first commercial robotic vacuum cleaner. Improved robustness of its visual navigation system in the areas of camera calibration, high dynamic range omni-directional imaging, illumination system, localisation and mapping system.
  - Delivered a high precision embedded visual-inertial odometry system for next-generation floor-care robots.
  - Delivered a motion tracking lab, from requirements specification to commissioning.
  - Integrated state-of-the-art academic research into commercial products.
  - Influenced large vendors to align their product development with the needs of our business.
  - Represented the business at trade shows and public events.

- **Senior Software Engineer** *Sep. 2012 – Sep. 2013*
  Guidance Navigation Limited, Leicester, United Kingdom

  - Transferred natural feature navigation technology from academia to industrial products.
  - Developed design specifications for driverless control systems for manual fork-lift trucks as per BS-EN 1525:1998 (Safety of Industrial Trucks. Driverless Trucks and Systems)
  - Modified mathematical formulation to improve precision of autonomous navigation systems.
  - Supported strategically important customers with on-site product testing and validation.

- **Senior Engineer** *Sep. 2011 – Sep. 2012*
  Blue Bear Systems Research (A Saab Company), Bedford, United Kingdom

  - Developed control and payload subsystems of unmanned aerial vehicles for urban ISTAR (Intelligence, Surveillance, Target acquisition and Reconnaissance) operations.
  - Worked with defense primes on the verification and validation of subsystems on the Long Endurance Multi-Intelligence Vehicle (LEMV).
  - Improved efficiency and safety of flight trials and operations by applying my experience as a qualified private pilot.
  - Demonstrated products at trade shows and public events.

- **Senior Systems Engineer** *Oct. 2006 – Sep. 2011*
  OC Robotics (GE Aerospace), Bristol, United Kingdom

  - Prototyped 3D vision systems for surface inspection, localisation and object identification. Specified hardware components such as imagers, optics, LCoS projectors, video processors, and lighting; delivered software for image analysis, machine vision and metrology.
  - Extended Linear Programming concepts to design a novel model-based gravity compensation algorithm that enhanced the controllability of slender Snake-arm robots in confined spaces.
  - Implemented human-machine interfaces (HMIs) for aerospace, nuclear and defense clients, each with unique functional and performance requirements.
  - Led a team that delivered the first version of *Snake Arm Simulator,* a professional simulation and analysis suite that demonstrated the unique capabilities of Snake-arm robots.
  - Proposed, designed and delivered a high-value simulator platform for SAFIRE, a bespoke Snake-arm robot to conduct inspections within the Upper Feeder Cabinets of CANDU nuclear reactors. The simulator was used to train inspection engineers, and for mission planning during reactor inspection outages.
  - Managed a high-value commercial proposal for robotic inspection of pressure vessels in an 'upstream' natural gas processing plant operated by an oil and gas ‘supermajor’. Defined the project scope, worked with senior management on risk assessments and costings, directed junior engineers in generating design options, and presented a comprehensive set of solutions to the customer. This included a proposal for an innovative software product for spatial and temporal geo-tagging of 3D environment models with visual inspection data (images, lidar scans, PDF reports, etc) to capture and archive inspection reports over time.
  - Advocated the use of documentation tools, industry standard software development guidelines (JSF AV rules, MISRA C++) and code reviews to ensure high quality standards. The team was audited routinely by external agents for quality and always met ISO/IEC 90003:2004 requirements.

- **Graduate Research Assistant**, *Mechatronics Laboratory* *Aug. 2000 – Aug. 2006*
  Clemson University, SC, USA

  - Led the development of electronics and control systems for the Clemson Octor biomimetic continuum robot with funding support from DARPA’s *Biodynotics* program.
  - Solved the challenge of shape measurement in continuum robots using cable-extension transducers (string potentiometers) used in automotive, aerospace and civil engineering industries. (http://cvilas.github.io/clemson-archive/projects/octor)
  - Key developer of the *Robotic Platform*, an open research and development platform for hard real-time robotic applications. The system integrated closed loop control, trajectory generation, advanced numerical computation and 3D visualization on a single PC – a significant achievement at the time. (http://cvilas.github.io/clemson-archive/projects/rp)
  - Developed new types of non-linear controllers and signal estimators by extending computer vision techniques with Lyapunov design methods. Applications included vision-based control of micro air vehicles, rigid body motion estimation, and structure-from-motion.
    - http://cvilas.github.io/clemson-archive/projects/sfm
    - http://cvilas.github.io/clemson-archive/projects/landing

Education

- **Emerging CTO Programme** *2026*
  Imperial College Business School (Executive Education)
- **Self Driving Car Engineer Nanodegree** *2018*
  Udacity
- **Ph.D.**, *Electrical Engineering (Focus: Robotics and Nonlinear Control)* *2006*
  Clemson University, USA
  GPA 3.8/4.0
- **M.S.**, *Electrical Engineering (Focus: Robotics and Nonlinear Control)* *2003*
  Clemson University, USA
  GPA 3.78/4.0
- **B.Eng.**, *Electrical and Electronics Engineering* *1999*
  University of Madras, India
  GPA 85.6/100 (University Silver Medallist)

Additional Qualifications

- Chartered Engineer (CEng), awarded by the Engineering Council, UK, Feb 2016.
- Senior Member, Institute of Electrical and Electronics Engineers (IEEE).
- Flight Crew Licence (expired): JAR-FCL PPL(A) SEP (Land), UK Civil Aviation Authority, June 2010.

Voluntary Service

Co-host, London.Robotics (2019 - 2021): See https://www.meetup.com/london-robotics/

Executive Team Member, IET Robotics and Mechatronics Network (2015 - 2019): Organised public lectures and networking events.

Reviewer (2002 - 2016): Served as a reviewer for academic publications including:
- IEEE/ASME Transactions on Mechatronics
- IEEE Transactions on Systems, Man and Cybernetics
- IEEE Transactions on Control Systems Technology
- IEEE Transactions on Robotics
- The International Journal of Robotics Research

Publications

- **Essays**
  - On the kind of colleagues I wish for.
  - On why homework assignments are a bad idea in recruitment. Review work samples instead.
  - On the challenges of offering robots as a service.

- **Book Chapters:**
  - D. Lee, T. C. Burg, V. Chitrakaran, E. Tatlicioglu, and C. Natraj, "Model-based 3D Position and Angle Trajectory-Tracking Control of Underactuated Unmanned Aerial Vehicle," *Numerical Simulation*, Academy Publishing, 2013.

- **Selected Journal Papers:**
  - J. Chen, V. K. Chitrakaran, and D. M. Dawson, "Range Identification of Features on an Object Using a Single Camera," *Automatica*, Vol. 47, No. 1, pp. 201 – 206, January 2011.
  - V. K. Chitrakaran, D. M. Dawson, W. E. Dixon, and J. Chen, "Identification of a Moving Object's Velocity with a Fixed Camera," *Automatica*, Vol. 41, No. 3, pp. 553 – 562, March 2005.
  - M. S. Loffler, V. K. Chitrakaran, and D. M. Dawson, "Design and Implementation of the Robotic Platform," *Journal of Intelligent and Robotic Systems*, Vol. 39, No. 1, pp. 105 – 129, January 2004.

- **Theses:**
  - Vilas K. Chitrakaran, "Lyapunov-Based Nonlinear Estimation and Control Using Vision in the Loop," *Ph.D. Thesis*, Dept. of Electrical and Computer Engineering, Clemson University, August 2006.
  - Vilas K. Chitrakaran, "The Robotic Platform – Implementation of the Manipulator Classes and the Math Library," *Master's Thesis*, Dept. of Electrical and Computer Engineering, Clemson University, December 2002.

- **Selected Conference Papers:**
  - V. K. Chitrakaran, and D. M. Dawson, "A Lyapunov-Based Method for Estimation of Euclidean Position of Static Features Using a Single Camera", *Proc. of the 26th American Control Conference*, pp. 1988-1993, July 2007.
  - W. McMahan, M. Pritts, V. Chitrakaran, D. Dienno, B. Jones, M. Grissom, M. Csencsits, V. Iyengar, I. Walker, C. Rahn, and D. Dawson, "Field Trials and Testing of the Octarm Continuum Manipulator," *2006 IEEE International Conference on Robotics and Automation*, pp. 2336-2341, May 2006.
  - V. K. Chitrakaran, D. M. Dawson, J. Chen, and M. Feemster, "Vision Assisted Autonomous Landing of an Unmanned Aerial Vehicle," *Proc. of the 44th IEEE Conference on Decision and Control*, pp. 1465-1470, Dec. 2005.